

Comparitive Performance Report for tentacle-3-way-replica-balance-reads-off vs direct-reads-3-way-replica-balance-reads- off-pr2-rerun

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Comparison summary for tentacle-3-way-replica-balance-reads-off vs direct-reads-3-way-replica-balance-reads- off-pr2-rerun

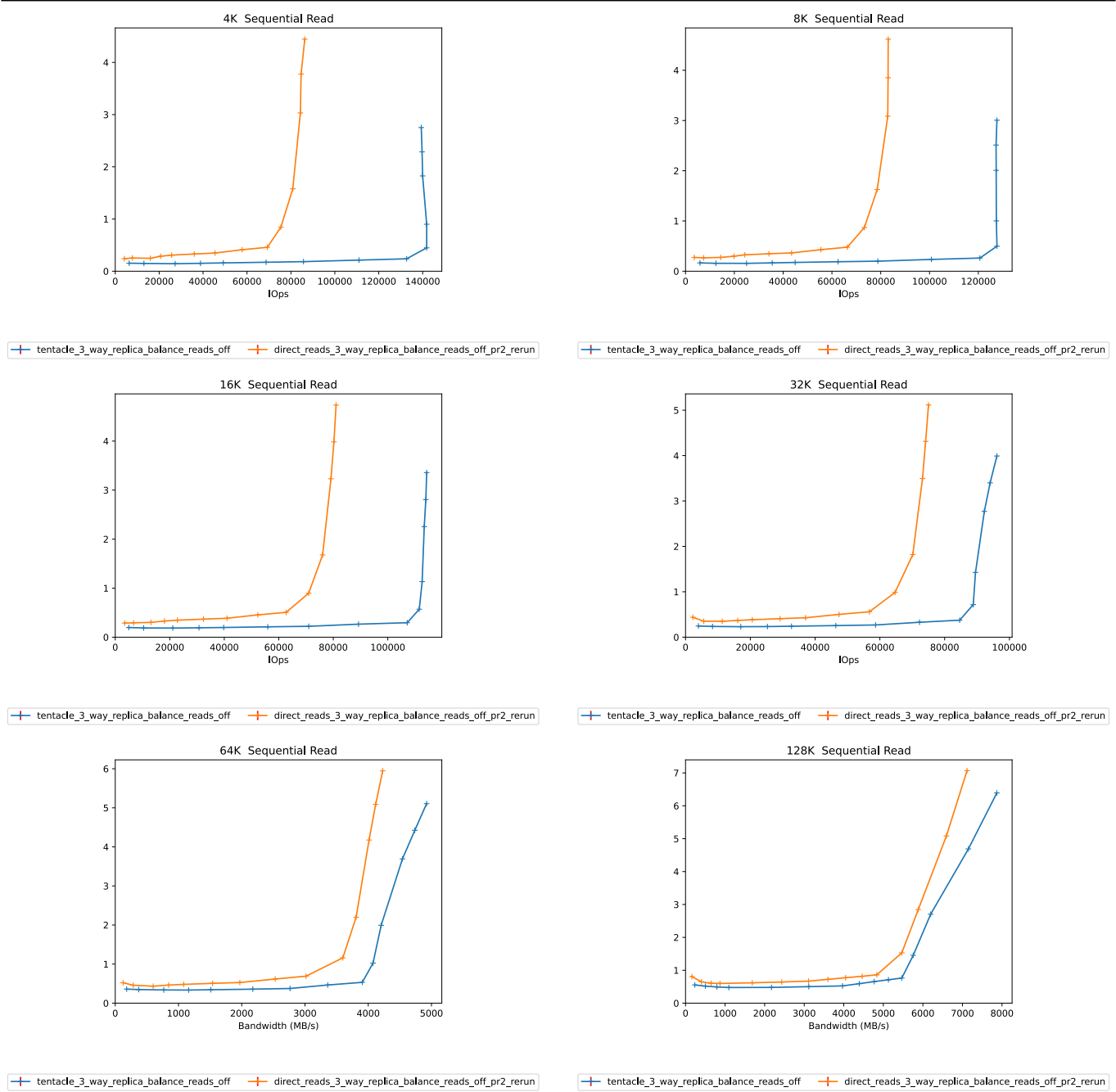
Sequential Read	tentacle_3_way_replica_balance_reads_off	direct_reads_3_way_replica_balance_reads_off_pr2_rerun	%change	latency
4K	141831 IOps@0.4ms	86307 IOps@4.4ms	-39%	1000%
8K	127635 IOps@0.5ms	83059 IOps@4.6ms	-35%	820%
16K	114277 IOps@3.4ms	81033 IOps@4.7ms	-29%	38%
32K	96131 IOps@4.0ms	75013 IOps@5.1ms	-22%	27%
64K	4923 MB/s@5.1ms	4229 MB/s@5.9ms	-14%	16%
128K	7869 MB/s@6.4ms	7116 MB/s@7.1ms	-10%	11%
256K	11495 MB/s@8.8ms	10785 MB/s@9.3ms	-6%	6%
384K	12087 MB/s@16.7ms	11635 MB/s@17.3ms	-4%	4%
512K	12738 MB/s@21.1ms	12690 MB/s@21.2ms	-0%	0%
768K	12620 MB/s@31.9ms	12926 MB/s@31.2ms	2%	-2%
1024K	12438 MB/s@10.8ms	13329 MB/s@30.2ms	7%	180%
2048K	11426 MB/s@11.7ms	12220 MB/s@11.0ms	7%	-6%
4096K	10350 MB/s@25.9ms	11286 MB/s@23.8ms	9%	-8%

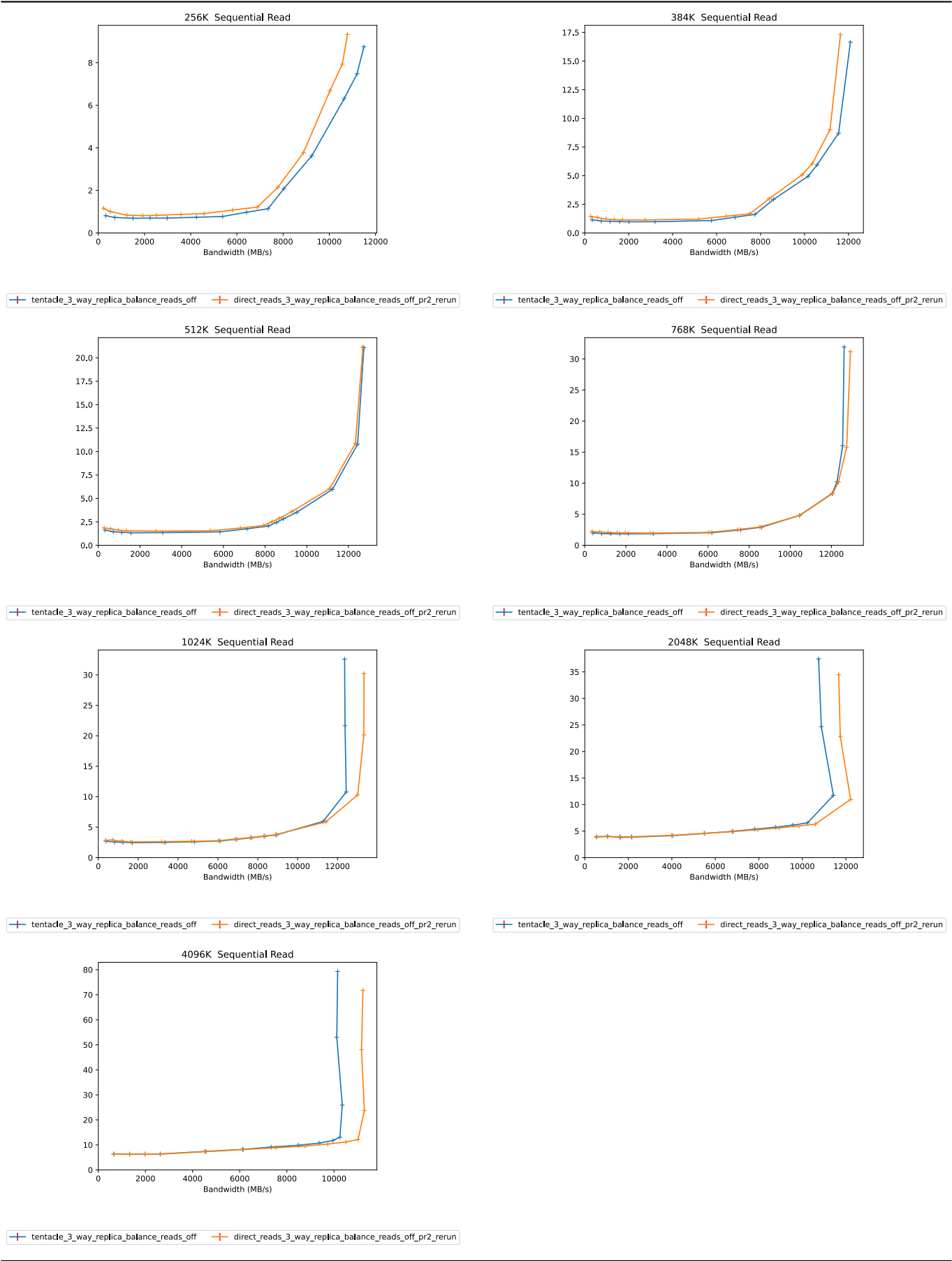
Random Read	tentacle_3_way_replica_balance_reads_off	direct_reads_3_way_replica_balance_reads_off_pr2_rerun	%change	latency
4K	153787 IOps@2.1ms	103584 IOps@3.7ms	-33%	76%
8K	146094 IOps@2.6ms	99601 IOps@3.2ms	-32%	23%
16K	137395 IOps@2.8ms	96017 IOps@4.0ms	-30%	43%
32K	116730 IOps@2.7ms	88133 IOps@4.3ms	-24%	59%
64K	6059 MB/s@2.8ms	4998 MB/s@5.0ms	-18%	79%
128K	9360 MB/s@3.6ms	8128 MB/s@6.2ms	-13%	72%
256K	12040 MB/s@7.0ms	10455 MB/s@8.0ms	-13%	14%
384K	12176 MB/s@4.1ms	12194 MB/s@8.2ms	0%	100%
512K	12548 MB/s@5.3ms	12981 MB/s@10.3ms	3%	94%
768K	11838 MB/s@4.2ms	13078 MB/s@7.7ms	10%	83%
1024K	12103 MB/s@5.5ms	12788 MB/s@10.5ms	6%	91%
2048K	11155 MB/s@12.0ms	12058 MB/s@11.1ms	8%	-8%
4096K	10266 MB/s@26.1ms	11218 MB/s@23.9ms	9%	-8%

Random Read/Write	tentacle_3_way_replica_balance_reads_off	direct_reads_3_way_replica_balance_reads_off_pr2_rerun	%change	latency
4K_70/30	82725 IOps@3.9ms	62792 IOps@6.1ms	-24%	56%
16K_70/30	75346 IOps@5.1ms	59158 IOps@6.5ms	-21%	27%
64K_30/70	2598 MB/s@8.0ms	2252 MB/s@11.1ms	-13%	39%
64K_70/30	3690 MB/s@6.8ms	3127 MB/s@5.3ms	-15%	-22%

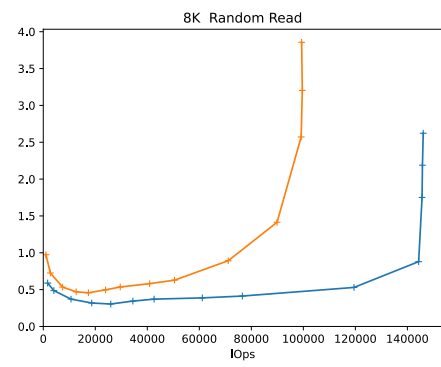
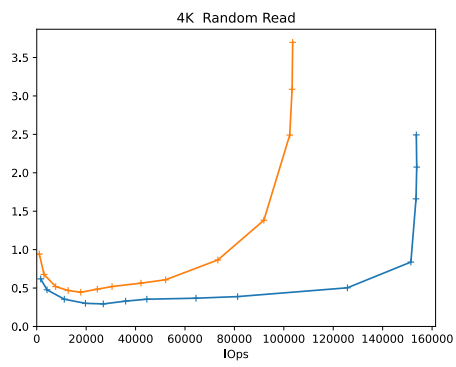
Response Curves

Sequential Read



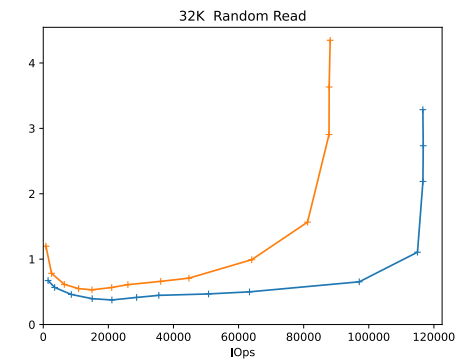
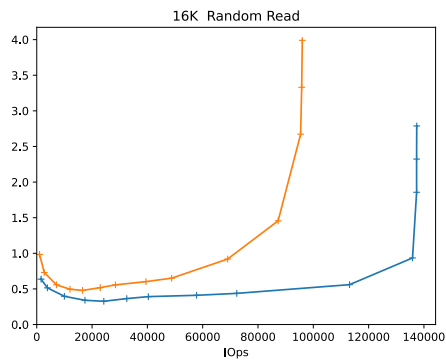


Random Read



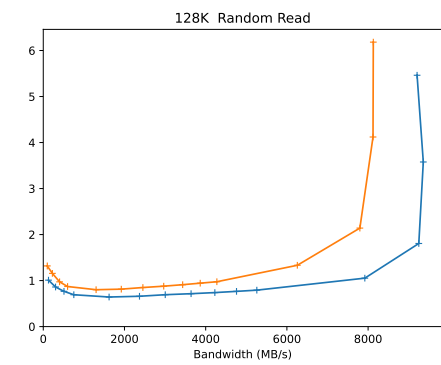
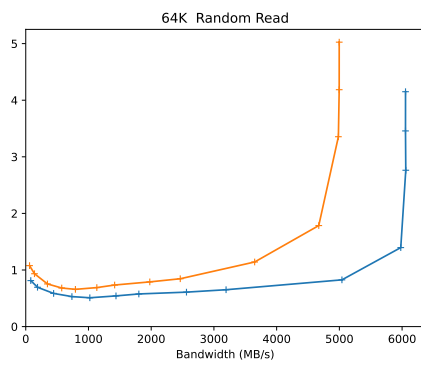
tentacle_3_way_replica_balance_reads_off direct_reads_3_way_replica_balance_reads_off_pr2_rerun

tentacle_3_way_replica_balance_reads_off direct_reads_3_way_replica_balance_reads_off_pr2_rerun



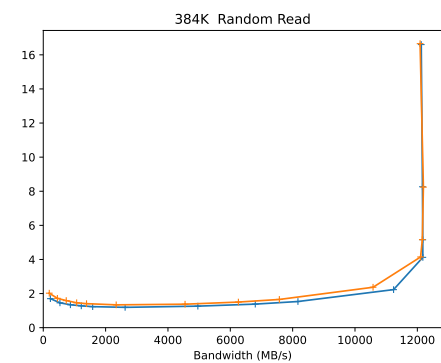
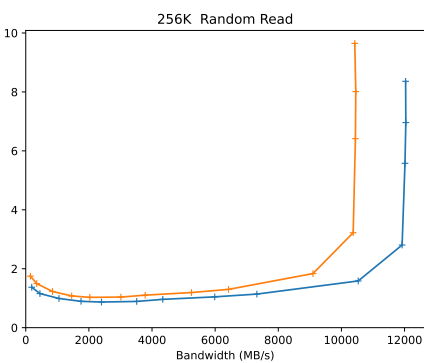
tentacle_3_way_replica_balance_reads_off direct_reads_3_way_replica_balance_reads_off_pr2_rerun

tentacle_3_way_replica_balance_reads_off direct_reads_3_way_replica_balance_reads_off_pr2_rerun



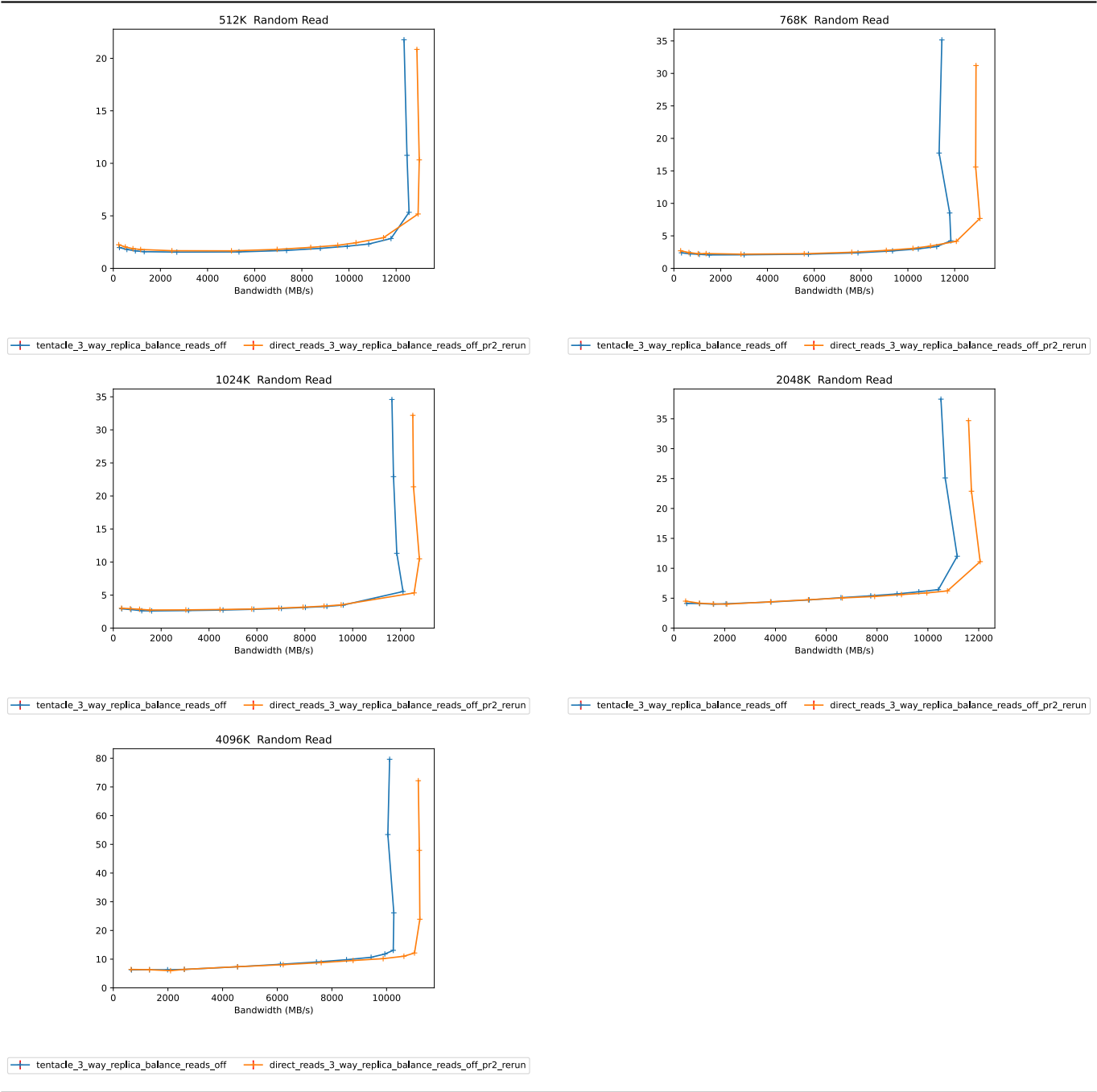
tentacle_3_way_replica_balance_reads_off direct_reads_3_way_replica_balance_reads_off_pr2_rerun

tentacle_3_way_replica_balance_reads_off direct_reads_3_way_replica_balance_reads_off_pr2_rerun

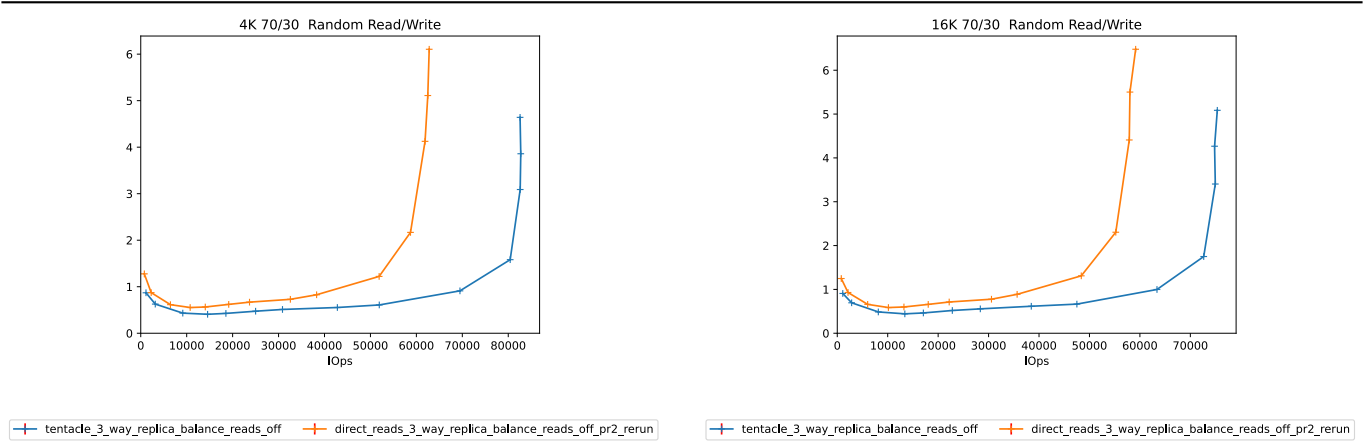


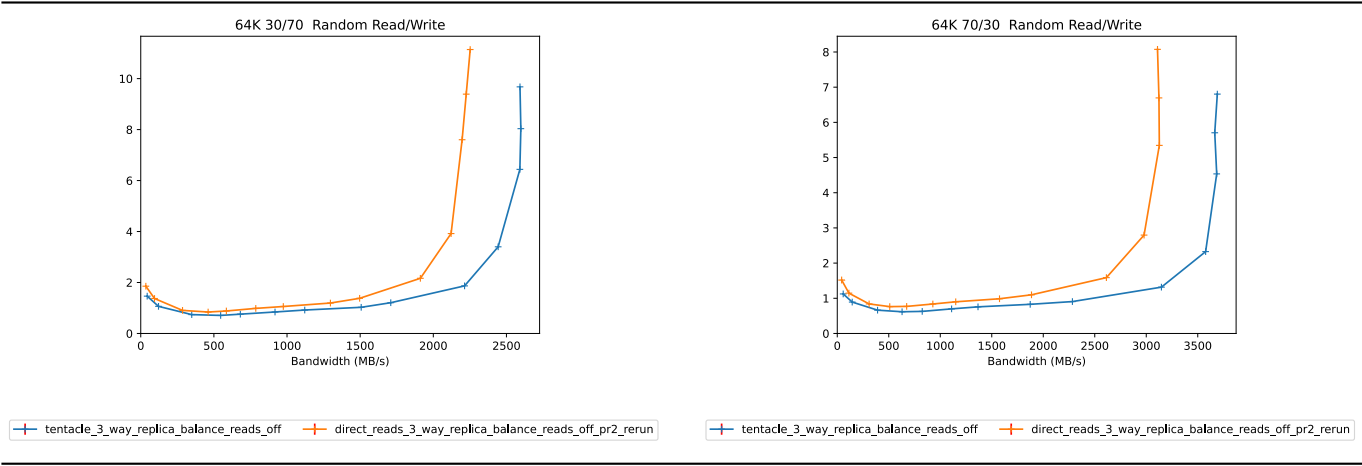
tentacle_3_way_replica_balance_reads_off direct_reads_3_way_replica_balance_reads_off_pr2_rerun

tentacle_3_way_replica_balance_reads_off direct_reads_3_way_replica_balance_reads_off_pr2_rerun



Random Read/Write





Configuration yaml files

Only yaml files that differ by more than 20 lines from the yaml file for the baseline directory will be added here in addition to the baseline yaml

results

```

librbd fio:
  cmd_path: /usr/local/bin/fio2
  create_report: true
  fio_out_format: json
  log_avg_msec: 100
  log_bw: true
  log_iops: true
  log_lat: true
  norandommap: true
  osd_ra:
    - 4096
  poolname: rbd_replicated
  prefill:
    blocksize: 64k
    numjobs: 1
  procs_per_volume:
    - 1
  ramp: 30
  rbdname: cbt-librbd fio
  time: 90
  time_based: true
  use_existing_volumes: true
  vol_size: 1000
  volumes_per_client:
    - 16
  wait_pgautoscaler_timeout: 10
  workloads:
    128k randomread:
      jobname: randread
      mode: randread
      numjobs:
        - 1
      op_size: 131072
      total_iodepth:
        - 1
        - 2
        - 3
        - 4
        - 8
        - 12
        - 16
        - 20
        - 24
        - 28
        - 32
        - 64
        - 128
        - 256
        - 384
    128k sequentialread:
      jobname: seqread
      mode: read
      numjobs:
        - 1
      op_size: 131072
      total_iodepth:
        - 1
        - 2
        - 3
        - 4
        - 8

```

```
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 256
- 384
16kmixread70:
  jobname: mixread
  mode: randrw
  numjobs:
    - 1
  op_size: 16384
  rwmixread: 70
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
16krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 16384
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
16ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 16384
  total_iodepth:
    - 1
    - 2
    - 4
```

```

- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384
1Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 1048576
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 256
    - 384
1Msequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 1048576
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 256
    - 384
256krandomread:
  jobname: randomread
  mode: randread
  numjobs:
    - 1
  op_size: 262144
  total_iodepth:
    - 1

```

```

- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384
256ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 262144
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
2Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 2097152
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 192
2Msequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 2097152
  total_iodepth:
    - 1

```

```

- 2
- 3
- 4
- 8
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 192
32krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 32768
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
32ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 32768
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
384krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 393216
  total_iodepth:
    - 1

```

```

- 2
- 3
- 4
- 5
- 8
- 16
- 24
- 32
- 64
- 128
- 160
- 256
- 512
384ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 393216
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 5
    - 8
    - 16
    - 24
    - 32
    - 64
    - 128
    - 160
    - 256
    - 512
4Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 4194304
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 192
4Msequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 4194304
  total_iodepth:
    - 1

```

```
- 2
- 3
- 4
- 8
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 192
4kmixread70:
  jobname: mixread
  mode: randrw
  numjobs:
    - 1
  op_size: 4096
  rwmixread: 70
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
4krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 4096
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
4ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 4096
  total_iodepth:
```



```

- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384
512krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 524288
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 16
    - 24
    - 32
    - 40
    - 48
    - 64
    - 128
    - 256
    - 512
512ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 524288
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 16
    - 24
    - 32
    - 40
    - 48
    - 64
    - 128
    - 256
    - 512
64kmixread30:
  jobname: mixread
  mode: randrw
  numjobs:
    - 1
  op_size: 65536
  rwmixread: 30

```

```
total_iodepth:
- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384
64kmixread70:
  jobname: mixread
  mode: randrw
  numjobs:
  - 1
  op_size: 65536
  rwmixread: 70
  total_iodepth:
  - 1
  - 2
  - 4
  - 6
  - 8
  - 12
  - 16
  - 24
  - 32
  - 64
  - 128
  - 256
  - 320
  - 384
64krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 65536
  total_iodepth:
  - 1
  - 2
  - 4
  - 6
  - 8
  - 12
  - 16
  - 24
  - 32
  - 64
  - 128
  - 256
  - 320
  - 384
64ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
  - 1
```

```

  op_size: 65536
  total_iodepth:
  - 1
  - 2
  - 4
  - 6
  - 8
  - 12
  - 16
  - 24
  - 32
  - 64
  - 128
  - 256
  - 320
  - 384
768krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 786432
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 16
  - 24
  - 32
  - 40
  - 48
  - 64
  - 128
  - 256
  - 512
768ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
  - 1
  op_size: 786432
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 5
  - 8
  - 16
  - 24
  - 32
  - 64
  - 128
  - 160
  - 256
  - 512
8krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1

```

```

    op_size: 8192
    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  8ksequentialread:
    jobname: seqread
    mode: read
    numjobs:
      - 1
    op_size: 8192
    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  precondition:
    jobname: precond1rw
    mode: randwrite
    monitor: false
    numjobs:
      - 1
    op_size: 65536
    time: 600
    total_iodepth:
      - 16
cluster:
  archive_dir: /tmp/cbt
  ceph_mgr_cmd: /usr/bin/ceph-mgr
  ceph_mon_cmd: /usr/bin/ceph-mon
  ceph_osd_cmd: /usr/bin/ceph-osd
  ceph_run_cmd: /usr/bin/ceph-run
  ceph_cmd: /usr/bin/ceph
  clients:
    - --- server1 ---
  clusterid: ceph
  conf_file: /cbt/ceph.conf.4x1x1.fs
  fs: xfs
  head: --- server1 ---
  iterations: 1
  mgrs:
    --- server1 ---:

```

```
    a: null
mkfs_opts: -f -i size=2048
mons:
  --- server1 ---:
    a: --- IP Address ---:6789
mount_opts: -o inode64,noatime,logbsize=256k
osds:
- --- server1 ---
  osds_per_node: 6
  pdsh_ssh_args: -a -x -l%u %h
  rados_cmd: /usr/bin/rados
  rbd_cmd: /usr/bin/rbd
  tmp_dir: /tmp/cbt
  use_existing: true
  user: cjames
monitoring_profiles:
  collectl:
    args: -c 18 -sCD -i 10 -P -oz -FO --rawtoo --sep ";" -f {collectl_dir}
```